MA087G Mathematical Statistics
Beginner-Friendly Full Solution Guide
2025-01-10

Source notes before solving

The official exam PDF was image-only. Normal text extraction found no text, but the rendered page images were readable.

Per request, this folder does not include an extraction log or mistake bank.

Uppgift 1

What this question is really asking

This is a three-set counting problem. There are 30000 students, and students can be active in areas A, B, and C.

We must find the probability that a randomly chosen student is active in:

- at least one area,
- no area,
- exactly one area.

How to recognize this type of question

The clues are:

- The question gives counts for A, B, and C.
- It gives overlaps like both A and B.
- It gives the number active in all three.

This is a Venn diagram/inclusion-exclusion problem.

Tiny theory needed

For three events:

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|.$$

The vertical bars mean "number of people in".

We subtract pair-overlaps because they were counted twice. We add the triple-overlap back because it was subtracted too many times.

Formula and what everything means

$$P(\text{event}) = \frac{\text{number of favorable students}}{\text{total number of students}}.$$

Here the total is 30000.

For exactly one area:

$$\text{only } A = |A| - |A \cap B| - |A \cap C| + |A \cap B \cap C|.$$

We add the triple back because it was removed twice.

Step-by-step solution

Given:

$$|A| = 12000, \quad |B| = 8000, \quad |C| = 6000.$$
$$|A \cap B| = 7000, \quad |A \cap C| = 4500, \quad |B \cap C| = 1000.$$
$$|A \cap B \cap C| = 500.$$

Part a: At least one area.

At least one means $A \cup B \cup C$. Use inclusion-exclusion:

$$\begin{aligned} |A \cup B \cup C| &= 12000 + 8000 + 6000 \\ &\quad - 7000 - 4500 - 1000 + 500 \\ &= 26000 - 12500 + 500 \\ &= 14000. \end{aligned}$$

So:

$$\begin{aligned} P(\text{at least one}) &= \frac{14000}{30000} \\ &= \frac{7}{15} \\ &\approx 0.467. \end{aligned}$$

Part b: No area.

No area is the complement of at least one area:

$$\begin{aligned} P(\text{no area}) &= 1 - P(\text{at least one}) \\ &= 1 - \frac{7}{15} \\ &= \frac{8}{15} \\ &\approx 0.533. \end{aligned}$$

Part c: Exactly one area.

Exactly one means only A, only B, or only C.

Only A:

$$\begin{aligned} |A \text{ only}| &= 12000 - 7000 - 4500 + 500 \\ &= 1000. \end{aligned}$$

Only B:

$$\begin{aligned} |B \text{ only}| &= 8000 - 7000 - 1000 + 500 \\ &= 500. \end{aligned}$$

Only C:

$$\begin{aligned} |C \text{ only}| &= 6000 - 4500 - 1000 + 500 \\ &= 1000. \end{aligned}$$

Total exactly one:

$$1000 + 500 + 1000 = 2500.$$

So:

$$\begin{aligned} P(\text{exactly one}) &= \frac{2500}{30000} \\ &= \frac{1}{12} \\ &\approx 0.0833. \end{aligned}$$

Answer

$$\boxed{P(\text{at least one}) = \frac{7}{15} \approx 0.467}$$

$$\boxed{P(\text{no area}) = \frac{8}{15} \approx 0.533}$$

$$\boxed{P(\text{exactly one}) = \frac{1}{12} \approx 0.0833}$$

What to memorize for the exam

For three overlapping groups, use:

$$\text{singles} - \text{pair overlaps} + \text{triple overlap}.$$

Common trap

Do not subtract the triple overlap permanently. People in all three are included in every pair-overlap, so they need to be added back once.

Repeated exam pattern

This is a new pattern so far: three-set Venn/inclusion-exclusion counting. It is related to the event-probability logic from 2026-01 Uppgift 1 and 2025-08 Uppgift 1.

Uppgift 2

What this question is really asking

Screw lengths are normally distributed:

$$X \sim N(1.8, 0.4).$$

A screw is usable if its length is less than 2 cm.

We must find:

- probability one screw is usable,
- probability all 10 screws in a box are usable,
- how many screws are needed so the chance of at least one usable screw is above 0.99,
- a one-sided hypothesis test for whether the mean is now larger than 1.8.

How to recognize this type of question

The clues are:

- It says normally distributed, so use normal standardization.
- Several screws in a box means multiply independent probabilities.
- "At least one" usually means use the complement "none".
- "Greater than 1.8" means a right-tailed hypothesis test.

Tiny theory needed

To use the standard normal table, convert:

$$z = \frac{x - \mu}{\sigma}.$$

Here z says how many standard deviations x is from the mean.

For at least one:

$$P(\text{at least one}) = 1 - P(\text{none}).$$

For a known-σ test:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

Formula and what everything means

$$z = \frac{x - \mu}{\sigma}$$

where:

- x is the boundary, here 2,
- $\mu = 1.8$ is the mean,
- $\sigma = 0.4$ is the standard deviation.

For all 10 usable:

$$P(\text{all 10 usable}) = p^{10},$$

where p is the probability one screw is usable.

Step-by-step solution

Part a: One screw usable.

Usable means:

$$X < 2.$$

Standardize:

$$\begin{aligned} z &= \frac{2 - 1.8}{0.4} \\ &= \frac{0.2}{0.4} \\ &= 0.5. \end{aligned}$$

So:

$$P(X < 2) = P(Z < 0.5) = \Phi(0.5) \approx 0.6915.$$

Part b: All 10 usable.

Let:

$$p = 0.6915.$$

Assuming screws are independent:

$$\begin{aligned} P(\text{all 10 usable}) &= p^{10} \\ &= 0.6915^{10} \\ &\approx 0.0250. \end{aligned}$$

Part c: At least one usable with probability above 0.99.

At least one usable:

$$1 - P(\text{none usable}).$$

The probability one screw is not usable is:

$$1 - p = 1 - 0.6915 = 0.3085.$$

For n screws:

$$P(\text{none usable}) = (0.3085)^n.$$

We want:

$$1 - (0.3085)^n > 0.99.$$

Move terms:

$$(0.3085)^n < 0.01.$$

Take logarithms:

$$n \ln(0.3085) < \ln(0.01).$$

Since $\ln(0.3085) < 0$, division flips the inequality:

$$\begin{aligned} n &> \frac{\ln(0.01)}{\ln(0.3085)} \\ &\approx 3.92. \end{aligned}$$

So the smallest whole number is:

$$n = 4.$$

Part d: Hypothesis test.

We test:

$$H_0 : \mu = 1.8, \qquad H_1 : \mu > 1.8.$$

Given:

$$n = 10, \qquad \bar{x} = 2.0, \qquad \sigma = 0.4.$$

Test statistic:

$$\begin{aligned} z &= \frac{2.0 - 1.8}{0.4/\sqrt{10}} \\ &= \frac{0.2}{0.1265} \\ &\approx 1.58. \end{aligned}$$

For a right-tailed 5% test:

$$z_{0.95} \approx 1.645.$$

Since:

$$1.58 < 1.645,$$

we do not reject H_0.

Answer

$$\boxed{P(\text{one screw usable}) \approx 0.6915}$$

$$\boxed{P(\text{all 10 usable}) \approx 0.0250}$$

$$\boxed{n = 4 \text{ screws are needed for probability above 0.99}}$$

$$\boxed{\text{Do not reject } H_0 \text{ at 5\%.}}$$

What to memorize for the exam

For normal probabilities, standardize:

$$z = \frac{x - \mu}{\sigma}.$$

For at least one, use:

$$1 - P(\text{none}).$$

Common trap

In the test, do not reject just because $\bar{x} = 2.0$ is bigger than 1.8. It must be big enough compared with the standard error.

Repeated exam pattern

The hypothesis test is the same known-σ, one-sided mean-test pattern as 2025-08 Uppgift 6 and 2025-06 Uppgift 6.

Uppgift 3

What this question is really asking

We are given:

$$f(x) = \begin{cases} ax^3, & 0 < x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

We need to normalize it, find the CDF, compute a probability, and find the density of:

$$Y = \ln X.$$

How to recognize this type of question

The clues are:

- It gives a density with an unknown constant a.
- It asks for F_X, so we integrate to get the CDF.
- It defines a new random variable $Y = \ln X$, so this is a transformation question.

Tiny theory needed

A density must integrate to 1.

For a transformation $Y = \ln X$, first find the CDF:

$$F_Y(y) = P(Y \leq y).$$

Then differentiate $F_Y(y)$ to get the density $f_Y(y)$.

Formula and what everything means

$$F_Y(y) = P(Y \leq y) = P(\ln X \leq y).$$

Since the exponential function reverses the log:

$$\ln X \leq y \quad \Longleftrightarrow \quad X \leq e^y.$$

Step-by-step solution

Part a: Find a.

$$\begin{aligned} \int_0^1 ax^3\,dx &= 1 \\ a\left[\frac{x^4}{4}\right]_0^1 &= 1 \\ \frac{a}{4} &= 1 \\ a &= 4. \end{aligned}$$

Part b: Find $F_X(x)$.

For $0 < x < 1$:

$$\begin{aligned} F_X(x) &= \int_0^x 4t^3\,dt \\ &= [t^4]_0^x \\ &= x^4. \end{aligned}$$

So:

$$F_X(x) = \begin{cases} 0, & x \leq 0, \\ x^4, & 0 < x < 1, \\ 1, & x \geq 1. \end{cases}$$

Part c: Calculate $P(|X| < 1/2)$.

Since X only takes values between 0 and 1, $|X| = X$. So:

$$\begin{aligned} P(|X| < 1/2) &= P(X < 1/2) \\ &= F_X(1/2) \\ &= \left(\frac{1}{2}\right)^4 \\ &= \frac{1}{16}. \end{aligned}$$

Part d: Density of $Y = \ln X$.

Since $0 < X < 1$, we know:

$$Y = \ln X < 0.$$

So Y has support $y < 0$.

For $y < 0$:

$$\begin{aligned} F_Y(y) &= P(Y \le y) \\ &= P(\ln X \le y) \\ &= P(X \le e^y) \\ &= F_X(e^y) \\ &= (e^y)^4 \\ &= e^{4y}. \end{aligned}$$

Now differentiate:

$$f_Y(y) = \frac{d}{dy} e^{4y} = 4e^{4y}.$$

Therefore:

$$f_Y(y) = \begin{cases} 4e^{4y}, & y < 0, \\ 0, & \text{otherwise.} \end{cases}$$

Answer

$$\boxed{a = 4}$$

$$\boxed{F_X(x) = \begin{cases} 0, & x \le 0, \\ x^4, & 0 < x < 1, \\ 1, & x \ge 1 \end{cases}}$$

$$\boxed{P(|X| < 1/2) = \frac{1}{16}}$$

$$\boxed{f_Y(y) = \begin{cases} 4e^{4y}, & y < 0, \\ 0, & \text{otherwise} \end{cases}}$$

What to memorize for the exam

For transformations, a beginner-safe method is:

$$F_Y(y) = P(Y \le y),$$

then solve the inequality back to X.

Common trap

Do not forget the support. Since $0 < X < 1$, $\ln X$ is always negative.

Repeated exam pattern

Parts a–c repeat the continuous density pattern from 2026-01 Uppgift 2, 2025-08 Uppgift 2, and 2025-06 Uppgift 2. Part d connects to 2025-06 Uppgift 3c, where a transformed variable's distribution was found.

Uppgift 4

What this question is really asking

X is uniformly distributed on $(-1, 1)$. That means all values between -1 and 1 are equally likely.

Y's distribution is symmetric around 0. That means:

$$P(Y \leq 0) = \frac{1}{2}.$$

Also X and Y are independent.

How to recognize this type of question

The clues are:

- "likformigt fördelad" means uniform distribution.
- "symmetrisk kring origo" means half the probability is below 0 and half above 0.
- $\max(X, Y) \leq 0$ means both $X \leq 0$ and $Y \leq 0$.

Tiny theory needed

For a uniform distribution on (a, b), probability is interval length divided by total length:

$$P(c < X < d) = \frac{d - c}{b - a}.$$

For $X \sim U(-1, 1)$:

$$E(X) = 0, \qquad V(X) = \frac{(1 - (-1))^2}{12} = \frac{1}{3}.$$

Formula and what everything means

For $W = 2X - 1$:

$$F_W(w) = P(W \leq w) = P(2X - 1 \leq w).$$

Then solve:

$$X \leq \frac{w + 1}{2}.$$

Step-by-step solution

Part a.

$$\begin{aligned} P(-0.1 < X < 0.2) &= \frac{0.2 - (-0.1)}{1 - (-1)} \\ &= \frac{0.3}{2} \\ &= 0.15. \end{aligned}$$

Part b.

$$\begin{aligned} E(2X-1) &= 2E(X)-1 \\ &= 2\cdot 0 - 1 \\ &= -1. \end{aligned}$$

Variance:

$$\begin{aligned} V(2X-1) &= 2^2 V(X) \\ &= 4\cdot \frac{1}{3} \\ &= \frac{4}{3}. \end{aligned}$$

Part c.

Let:

$$W = 2X - 1.$$

Since $-1 < X < 1$, the new variable satisfies:

$$-3 < W < 1.$$

For $-3 < w < 1$:

$$\begin{aligned} F_W(w) &= P(W \le w) \\ &= P(2X-1 \le w) \\ &= P\left(X \le \frac{w+1}{2}\right). \end{aligned}$$

The CDF of $X \sim U(-1,1)$ is:

$$F_X(x) = \frac{x+1}{2}, \qquad -1 < x < 1.$$

Insert $x = (w+1)/2$:

$$\begin{aligned} F_W(w) &= \frac{\frac{w+1}{2}+1}{2} \\ &= \frac{w+3}{4}. \end{aligned}$$

So:

$$F_W(w) = \begin{cases} 0, & w \le -3, \\ \dfrac{w+3}{4}, & -3 < w < 1, \\ 1, & w \ge 1. \end{cases}$$

Part d.

$$\max(X,Y) \le 0$$

means:

$$X \le 0 \text{ and } Y \le 0.$$

By independence:

$$\begin{aligned} P(\max(X,Y) \le 0) &= P(X \le 0)P(Y \le 0) \\ &= \frac{1}{2}\cdot\frac{1}{2} \\ &= \frac{1}{4}. \end{aligned}$$

Answer

$$\boxed{P(-0.1 < X < 0.2) = 0.15}$$

$$\boxed{E(2X-1) = -1, \qquad V(2X-1) = \frac{4}{3}}$$

$$\boxed{F_{2X-1}(w) = \begin{cases} 0, & w \le -3, \\ \frac{w+3}{4}, & -3 < w < 1, \\ 1, & w \ge 1 \end{cases}}$$

$$\boxed{P(\max(X,Y) \le 0) = \frac{1}{4}}$$

What to memorize for the exam

Uniform probability is length over total length. For max questions:

$$\max(X,Y) \le c \iff X \le c \text{ and } Y \le c.$$

Common trap

For variance, the constant -1 does not matter, but the coefficient 2 gets squared:

$$V(2X-1) = 4V(X).$$

Repeated exam pattern

Part c repeats the transformed-variable CDF pattern from 2025-06 Uppgift 3c. Part d repeats the maximum pattern from 2025-06 Uppgift 3d and 2025-08 Uppgift 3b.

Uppgift 5

What this question is really asking

We have a density depending on θ:

$$f_X(x) = \theta(1+x)^{-1-\theta}, \qquad x \ge 0.$$

We know θ is only allowed to be:

$$2, \quad 3, \quad 4.$$

The sample values are:

$$0.2, \quad 0.8.$$

We must choose which θ makes the sample most likely.

How to recognize this type of question

The clue is "ML-skattningen". Usually we derive by differentiating, but here θ can only be 2, 3, or 4. So the easiest method is to calculate the likelihood for each and choose the largest.

Tiny theory needed

Likelihood means:

$$L(\theta) = f(0.2)f(0.8).$$

The ML estimate is the θ that gives the largest $L(\theta)$.

Formula and what everything means

$$L(\theta) = \theta(1+0.2)^{-1-\theta} \cdot \theta(1+0.8)^{-1-\theta}.$$

This becomes:

$$L(\theta) = \theta^2(1.2)^{-1-\theta}(1.8)^{-1-\theta}.$$

Step-by-step solution

Since:

$$1.2 \cdot 1.8 = 2.16,$$

we can write:

$$L(\theta) = \theta^2(2.16)^{-1-\theta}.$$

Now test the allowed values.

For $\theta = 2$:

$$L(2) = 2^2(2.16)^{-3} \approx 0.3969.$$

For $\theta = 3$:

$$L(3) = 3^2(2.16)^{-4} \approx 0.4135.$$

For $\theta = 4$:

$$L(4) = 4^2(2.16)^{-5} \approx 0.3403.$$

The largest likelihood is:

$$L(3) \approx 0.4135.$$

Therefore:

$$\hat{\theta}_{ML} = 3.$$

Answer

$$\boxed{\hat{\theta}_{ML} = 3}$$

What to memorize for the exam

If the parameter is only allowed to take a few values, you can simply calculate the likelihood for each value and choose the biggest.

Common trap

Do not give the continuous maximizer if the question says θ is either 2, 3, or 4. The answer must be one of those three.

Repeated exam pattern

This is the same ML pattern as 2026-01 Uppgift 5, 2025-08 Uppgift 4, and 2025-06 Uppgift 4, but with a discrete set of possible parameter values.

Uppgift 6

What this question is really asking

We have two groups of students and three possible grades:

$$F, \quad G, \quad VG.$$

We test whether the grade distribution is the same in both groups.

This is a chi-square test for homogeneity.

How to recognize this type of question

The clues are:

- Data are in a table of counts.
- There are two groups and several categories.
- The question asks whether the distributions are the same.

Tiny theory needed

The chi-square statistic is:

$$\chi^2 = \sum \frac{(O-E)^2}{E}.$$

Here:

- O is observed count,
- E is expected count if the distributions were the same.

Expected count:

$$E = \frac{\text{row total} \cdot \text{column total}}{\text{grand total}}.$$

Formula and what everything means

Degrees of freedom:

$$(r-1)(c-1).$$

Here:

- $r = 2$ rows,

- $c = 3$ grade categories.

So:

$$(2-1)(3-1) = 2.$$

Step-by-step solution

Observed table:

	F	G	VG	row total
group 1	15	10	5	30
group 2	23	27	10	60
column total	38	37	15	90

Hypotheses:

$$H_0 : \text{grade distribution is the same in both groups}$$

$$H_1 : \text{grade distribution is not the same in both groups.}$$

Expected counts for group 1:

$$E_{1,F} = \frac{30 \cdot 38}{90} = 12.667,$$

$$E_{1,G} = \frac{30 \cdot 37}{90} = 12.333,$$

$$E_{1,VG} = \frac{30 \cdot 15}{90} = 5.$$

Expected counts for group 2:

$$E_{2,F} = \frac{60 \cdot 38}{90} = 25.333,$$

$$E_{2,G} = \frac{60 \cdot 37}{90} = 24.667,$$

$$E_{2,VG} = \frac{60 \cdot 15}{90} = 10.$$

Now compute the statistic:

$$\begin{aligned}\chi^2 &= \frac{(15-12.667)^2}{12.667} + \frac{(10-12.333)^2}{12.333} + \frac{(5-5)^2}{5} \\ &\quad + \frac{(23-25.333)^2}{25.333} + \frac{(27-24.667)^2}{24.667} + \frac{(10-10)^2}{10} \\ &\approx 1.307.\end{aligned}$$

At 5% significance with 2 degrees of freedom:

$$\chi^2_{0.95,2} \approx 5.991.$$

Since:

$$1.307 < 5.991,$$

we do not reject H_0.

Answer

$$\boxed{\chi^2 \approx 1.307}$$

$$\boxed{\text{Do not reject } H_0.}$$

There is not enough evidence at the 5% level to say that the grade distribution differs between the two groups.

What to memorize for the exam

For a table of counts:

$$E = \frac{\text{row total} \cdot \text{column total}}{\text{grand total}}, \qquad \chi^2 = \sum \frac{(O-E)^2}{E}.$$

Common trap

Do not compare percentages by eye only. The chi-square test measures whether the differences are large relative to the expected counts.

Repeated exam pattern

This is a new pattern in the processed folders: chi-square test for homogeneity in a contingency table.

Mini-cheatsheet from this exam

Venn diagrams

$$|A \cup B \cup C| = |A| + |B| + |C| - |AB| - |AC| - |BC| + |ABC|.$$

Normal distribution

$$z = \frac{x - \mu}{\sigma}.$$

Known-σ test:

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}.$$

At least one

$$P(\text{at least one}) = 1 - P(\text{none}).$$

Density and transformations

$$\int f(x)\,dx = 1, \qquad F_X(x) = \int_{-\infty}^{x} f(t)\,dt.$$

For transformations:

$$F_Y(y) = P(Y \leq y).$$

Uniform distribution

For $X \sim U(a, b)$:

$$P(c < X < d) = \frac{d-c}{b-a}, \qquad V(X) = \frac{(b-a)^2}{12}.$$

Maximum

$$\max(X, Y) \leq c \iff X \leq c \text{ and } Y \leq c.$$

Maximum likelihood

If only a few parameter values are allowed, calculate L for each and choose the largest.

Chi-square table test

$$E = \frac{\text{row total} \cdot \text{column total}}{\text{grand total}}, \qquad \chi^2 = \sum \frac{(O-E)^2}{E}.$$